

Computer Science & IT

Discrete Mathematics



Mathematical Logic

Lecture No. 04



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Recap of Previous Lecture



Topic

Practice questions



Topic

Proof by contradiction



Topics to be Covered



Topic

Proof by contradiction

Topic

Conditional proof rule

Topic

Some important propositional statements

Topic

Predicate

Topic

Quantifiers

Topic

Scope of the quantifier

Q: Check whether the following argument is valid or invalid

$\{P \rightarrow Q, \sim(P \wedge Q)\}$ logically implies $\sim P$

Solution using Proof by Contradiction

① Assume that the Conclusion of the argument [i.e. $\sim P$] is false [i.e. Assume that $\sim P$ is false]

i.e. assume that P is true

② Add that false Conclusion in the set of premises.

∴ New set of premises becomes.

$\{P \rightarrow Q, \sim(P \wedge Q), P\}$

↓ MP

Q

$\sim P \vee \sim Q$

Disjunctive Syllogism

∴ $\sim Q$

it is a Contradiction ∴ our assumption is false
Hence the given argument is Valid

Q: Check whether the following argument is valid or invalid

$\{P \rightarrow Q, \sim(P \wedge Q)\}$ logically implies Q

After including the false conclusion into the set of Premises, new set of premises will be

$\{P \rightarrow Q, \sim(P \wedge Q), \sim Q\}$

Modus
Tollens

$\therefore \sim P$

$\sim P \vee \sim Q$

$\sim Q$

No Contradiction
 $\therefore$ given argument
is invalid

Q:- Check whether the following argument is valid or not?

$\{P \rightarrow (Q \rightarrow S), \sim R \vee P, Q\}$ logically implies " $R \rightarrow S$ "

Solution
Using proof
by Contradiction: $\rightarrow$

New set of premises will be

$\{P \rightarrow (Q \rightarrow S), \sim R \vee P, Q, \sim(R \rightarrow S)\}$



it is contradiction
Hence given argument is valid



Topic : Conditional proof

The argument,

$\{P_1, P_2, P_3, \dots, P_n\} \vdash Q \rightarrow R$ is valid

if and only if

the argument

$\{P_1, P_2, P_3, \dots, P_n, Q\} \vdash R$ is valid

i.e., Checking whether the argument $\{P_1, P_2, P_3, \dots, P_n\} \vdash Q \rightarrow R$ is valid or invalid is Exactly same as Checking for the validity of the argument $\{P_1, P_2, P_3, \dots, P_n, Q\} \vdash R$

Q:- Check whether the following argument is valid or not?

$\{P \rightarrow (Q \rightarrow S), \sim R \vee P, Q\}$ logically implies " $R \rightarrow S$ "

Solution using Conditional proof rule

$\{P \rightarrow (Q \rightarrow S), \sim R \vee P, Q\}$ logically implies " $R \rightarrow S$ "

if and only if

$\{P \rightarrow (Q \rightarrow S), \sim R \vee P, Q, R\}$ logically implies S

$\downarrow$ M.P.

$\therefore Q \rightarrow S$

$\therefore P \vee S$

M.P.
 $\therefore S$

$\therefore$ given implication is valid



Topic : Some important statements

1. $P \rightarrow Q \equiv \{ Q \text{ follows from } P \}$
2. $P \text{ implies } Q \equiv P \rightarrow Q$
3. $\text{If } P \text{ then } Q \equiv P \rightarrow Q$
4. $P \text{ only if } Q \equiv P \rightarrow Q$
5. $P \text{ is sufficient condition for } Q \equiv P \rightarrow Q$
6. $Q \text{ is necessary condition for } P \equiv P \rightarrow Q$

$P \rightarrow Q$ $\equiv$ you can clear the gate exam
only if you appear for the gate exam

① $\rightarrow$ if you cleared the gate exam
 P then you appeared for the gate exam
 $Q \equiv P \rightarrow Q$

② $\rightarrow$ If you appear for the gate exam
then you clear the gate exam
 $\equiv Q \rightarrow P$



Topic : Some important statements

7. $P \text{ if } Q \equiv Q \rightarrow P$

8. $P \text{ when } Q \equiv Q \rightarrow P$

9. $P \text{ follows from } Q \equiv Q \rightarrow P$

10. $P \text{ unless } \sim Q \equiv \sim(\sim Q) \rightarrow P$

to get the equivalent
propositional formula
simply replace unless by disjunction ($\vee$)

$$\equiv Q \rightarrow P \equiv \sim P \rightarrow \sim Q$$

i.e. $P \text{ unless } \sim Q \equiv P \vee \sim Q \equiv \sim Q \vee P \equiv Q \rightarrow P$
 $\equiv \sim P \rightarrow \sim Q$

eg:- you can not crack "gate" unless you appear for "gate"

ie, if you do not appear for the gate exam
then you can not crack the gate exam } $\equiv \sim Q \rightarrow P$

(or) if you have cracked gate exam
then you must have appeared for gate } $\equiv \sim P \rightarrow Q$

||| } Contradiction
at
each other

Predicate Logic

(First order predicate logic)



Topic : Predicate

It is part of a sentence or clause stating something about the subject.

eg: Ram is a Politician
Subject Predicate

{ The statement,
may be true
or may be false }

let, 'P' represent "is a politician"

$P(\text{Ram})$: Ram is a politician

Predicate 'P'
is applied over the Subject Ram

→ Let 'S' represent the predicate "is a sportsman"

$S(\text{Ram})$: Ram is a sportsman

$S(\text{Mohan})$: Mohan is a sportsman

{ Statements
may be true
or may be false

P : is a politician
 S : is a sportsman

$P(x)$: x is a politician
 $S(y)$: y is a sportsman

} x and y may be any subject from the set of people under consideration

$P(x) \rightarrow S(x)$: if " x is a politician" then " x is a sportsman"

it may be true
it may be false =

① if x is not a politician then implication will always return true, does not matter x is a sportsman or not

② If x is a politician then x must be sportsman as well, for implication to return true

③ If x is a politician, but x is not a sportsman then only the implication will return false

P : is a politician

S : is a sportsman

$P(x) \rightarrow S(y)$: if x is a politician then y is a sportsman
for implication to be true,
either ' x ' should not be a politician
or ' y ' must be a sportsman

$P(x) \vee S(y)$: x is a politician or y is a sportsman

$P(x) \wedge S(x)$: x is a politician cum sportsman

$P(x) \longleftrightarrow S(y)$: x is a politician if and only if y is a sportsman



Topic : Predicate with multiple subjects

Let 'F' represent " is a friend of "

$F(\text{Ram}, \text{Mohan})$: Ram is a friend of Mohan

it may be true, it may be false

$F(x, y)$: x is a friend of y

G : is greater than

$G(a, b)$: a is greater than b

$G(2, 3)$ is false

$G(7, 1)$ is true

$F(x, y, t)$: Person x can fool person y at time t



Topic : Quantifiers



- In predicate logic, predicates are used alongside quantifiers to express the extent to which a predicate is true over a range of elements.

i.e. over the elements of domain

(or)

over the elements of universe of discourse

- There are two types of quantifiers

1. Universal Quantifier $\forall$ { for all }

2. Existential Quantifier $\exists$ { for some
for ^{or} at least '1' }



Topic : Universal Quantifier ($\forall$)

i.e. for all 'x' in our domain 

$\forall$: for all

$\forall x$: for all x in the universe of discourse

let, 'S' : "is a sportsman"

And let's assume domain is set of all students in our class

$\forall x \{ S(x) \}$: All are sportsman

the statement is trying to say that Everyone in our class is a sportsman

Truth value of this predicate formula will be the truth value returned by quantifier for all ($\forall$)

it may be true or it may be false

- $\forall x \{S(x)\}$ will return false if and only if there is at least one element in the universe of discourse (i.e., in the domain) for which the Predicate $S(x)$ returns false.

Note: If universe of discourse (i.e., domain) happens to be empty, then quantifier " $\forall$ " will always return true, because there is no element for which the underlying Predicate can return false.

→ $\forall x \{S(x)\}$ will return true
if and only if,

① domain is Empty

(or) ② Predicate 'S' returns true for all the
elements in the domain

In $\forall x \{S(x)\}$

quantifier

Scope of the
quantifier

H.W. What exactly will be represented by following two Predicate formula.

$$(i) \quad \forall x \{P(x)\} \longrightarrow \forall x \{S(x)\}$$

$$(ii) \quad \forall x \{P(x) \longrightarrow S(x)\}$$

H.W. Which of the following Predicate formula is/are valid

$$F_1: \left[\forall x \{P(x)\} \longrightarrow \forall x \{S(x)\} \right] \longrightarrow \left[\forall x \{P(x) \longrightarrow S(x)\} \right]$$

$$F_2: \left[\forall x \{P(x) \longrightarrow S(x)\} \right] \longrightarrow \left[\forall x \{P(x)\} \longrightarrow \forall x \{S(x)\} \right]$$



2 mins Summary



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THANK - YOU